

Characteristics of Forces — Paper B

MEDIUM • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every answer must carry a unit, and every resultant force must carry a direction. An answer with no unit does not score the final mark.
- The mark for each question is shown in square brackets on the right.
- Reminder: weight = mass $\times$ gravitational field strength ($W = mg$), and on Earth $g = 10 \text{ N/kg}$.

SECTION 1 – DRAWING AND MEASURING FORCES

- 1.** A crate rests on the floor. In the space below, draw the crate and label the two vertical forces acting on it, using arrows. Then state what you notice about the lengths of your two arrows, and why. [4]

- 2.** Explain why a force is drawn as an arrow on a diagram, rather than written as a number on its own. [3]

- 3.** A newtonmeter is used to measure a force.
- (a)** State the unit that a newtonmeter reads in. [1]

(b) Name the part inside a newtonmeter that stretches. [1]

(c) State what happens to the stretch when the force pulling on it is doubled. [1]

SECTION 2 – RESULTANTS AND MOTION

4. Calculate the resultant force in each case. Give a direction as well as a size.

(a) 80 N to the right and 50 N to the right. [1]

(b) 90 N to the left and 35 N to the right. [1]

(c) 120 N to the right and 120 N to the left. [1]

(d) 6 N to the right, 14 N to the left and 4 N to the right. [2]

5. A lorry drives along a straight motorway at a steady 25 m/s. The driving force from its engine is 4500 N.

(a) State the size of the total backward force acting on the lorry. [1]

(b) Explain how you know. [1]

- (c)** The driver presses the accelerator and the driving force rises to 5200 N. [2]
Calculate the new resultant force and describe what happens to the lorry.

- 6.** Explain the difference between balanced and unbalanced forces, and state what each does to an object that is already moving. [3]

SECTION 3 – FRICTION

- 7.** Friction acts on a moving object.

- (a)** State the direction in which friction acts. [1]

- (b)** Give two everyday situations in which friction is useful. [2]

- (c)** Give one situation in which friction is a nuisance. [1]

- 8.** A cyclist stops pedalling on a flat road and gradually comes to a stop.

- (a)** Name the two forces that slow her down. [2]

- (b)** Explain, in terms of the resultant force on her, why she slows down. [2]

End of paper. Check every answer has a unit or a form the question actually asked for.